Multivariate analysis

UNIVERSIDAD ICESI | A OTRO NIVEL

Agenda

- Visual Análisis
- Python graphs

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Types of analysis

Numeric vs. Categorical (e.g. Price vs. fuel)

- Multiple histograms
- Multiple density estimate plots
- Multiple box plots

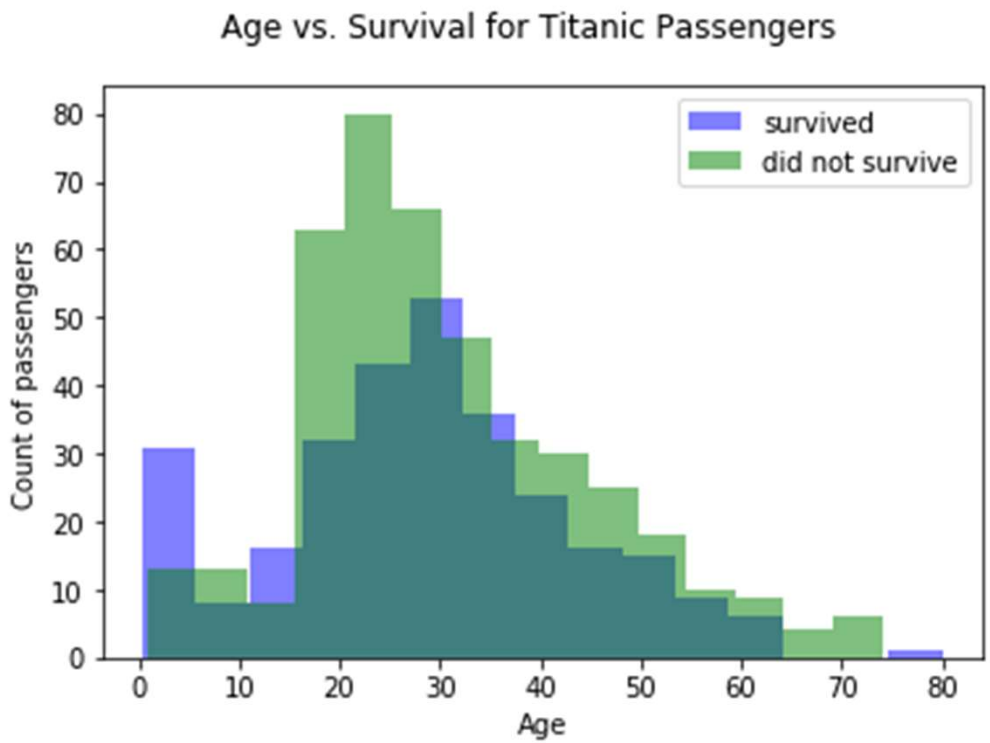
Categorical vs. Categorical (e.g. fuel vs. Chasis)

- Grouped bar charts
- Stacked bar charts

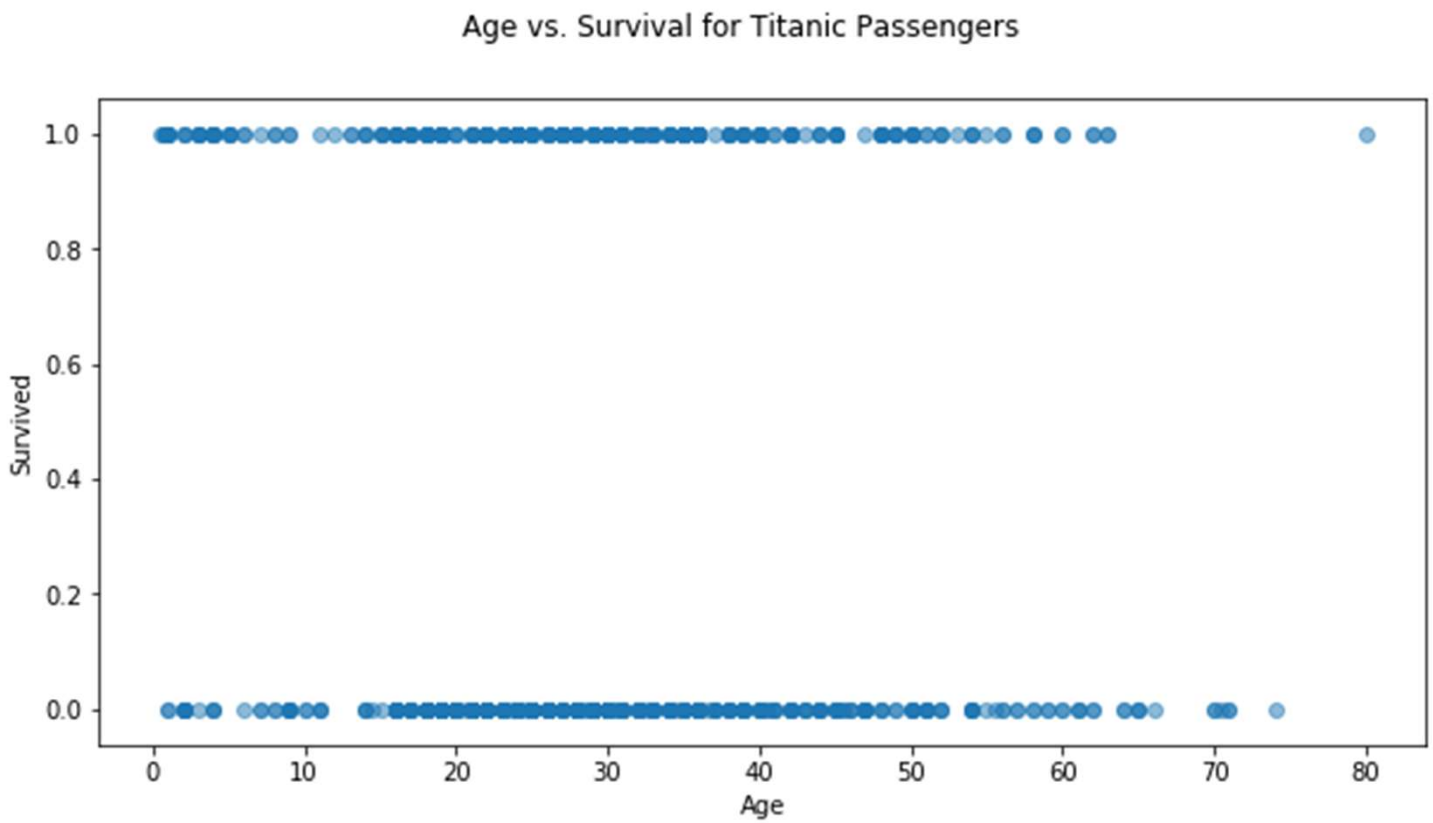
Numeric vs. Numeric vs. Categorical (e.g. price vs. Kms vs. type)

- Color-coded scatter plots

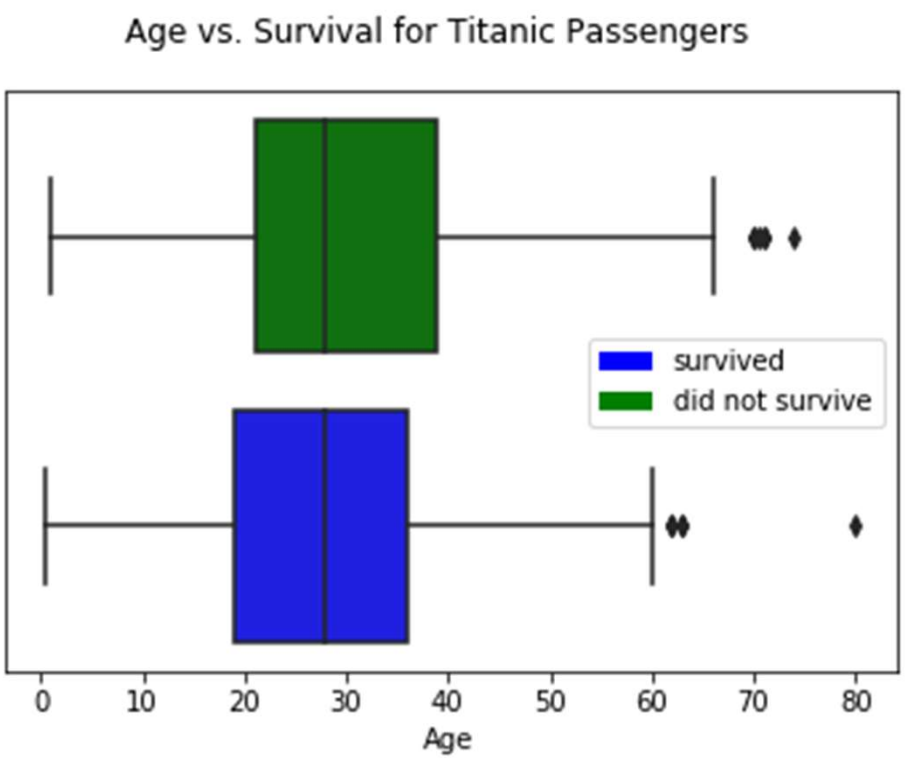
Numeric vs. Categorical



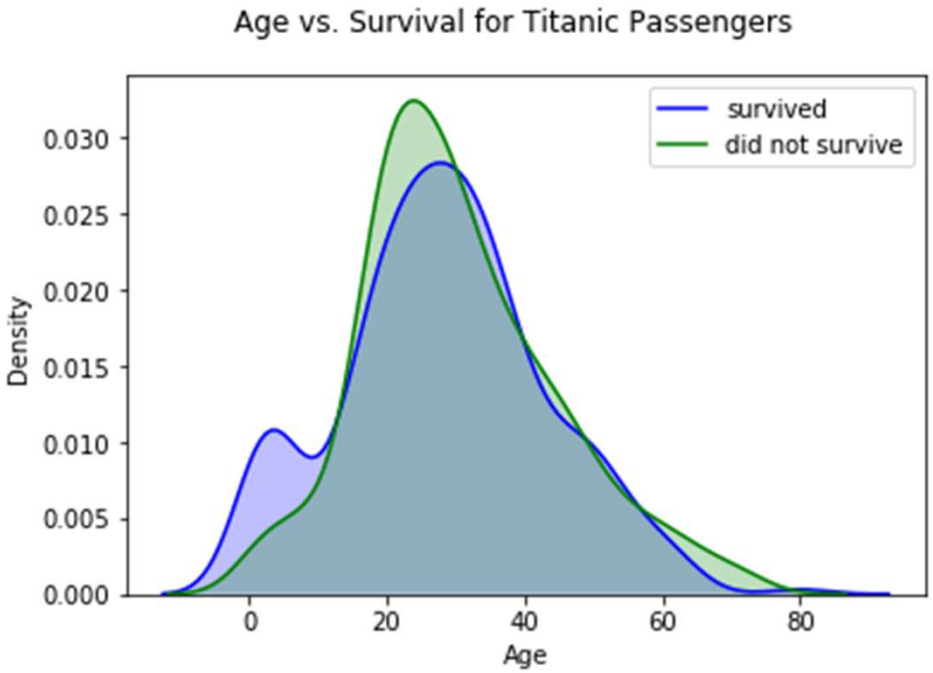
Numeric vs. Categorical



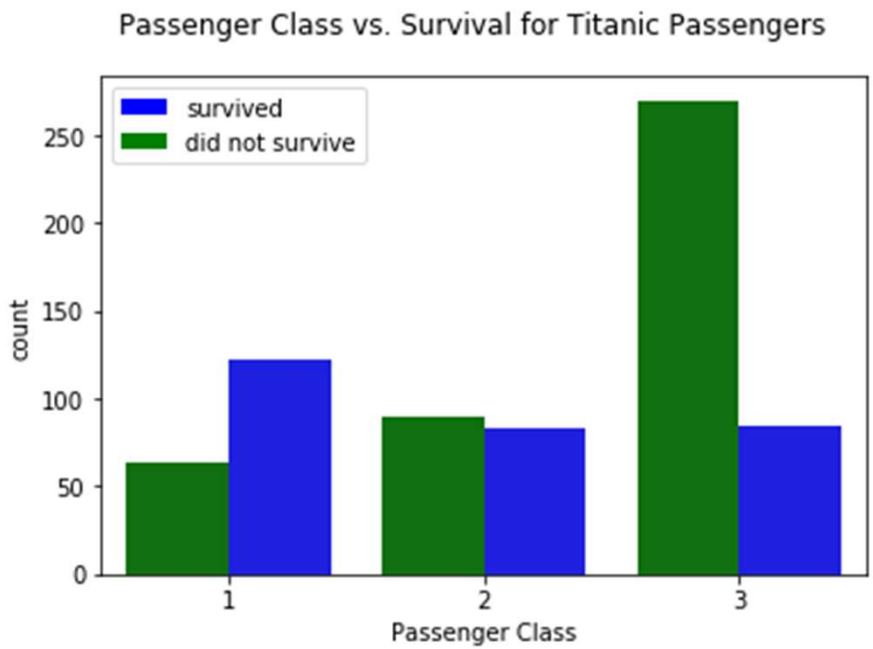
Multiple Box Plots



Numeric vs. Categorical

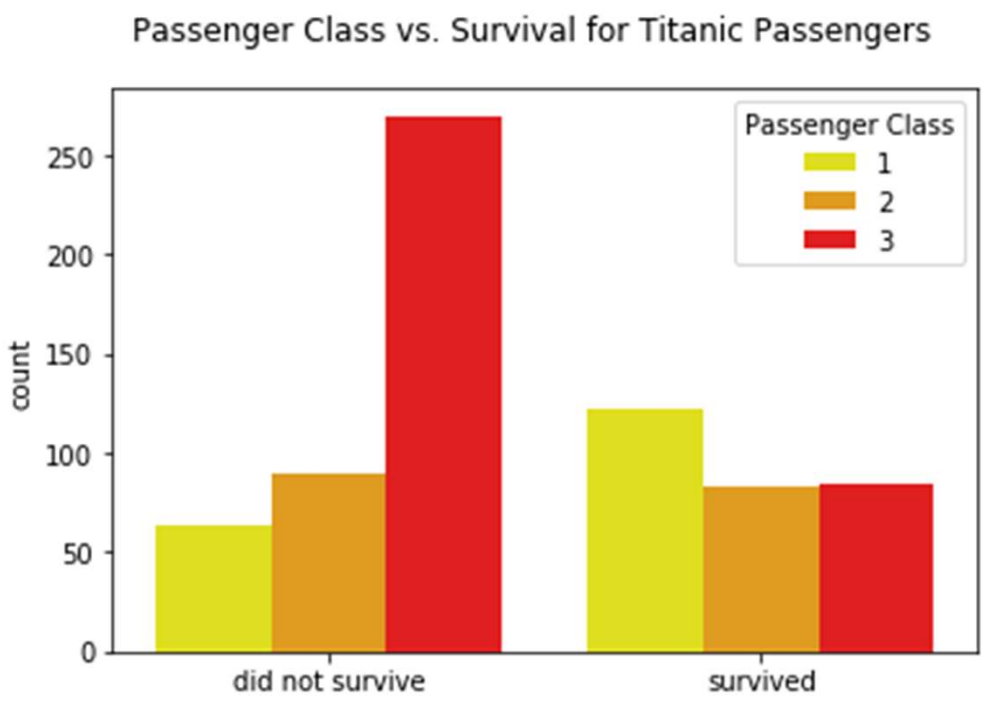


Categorical vs Categorical



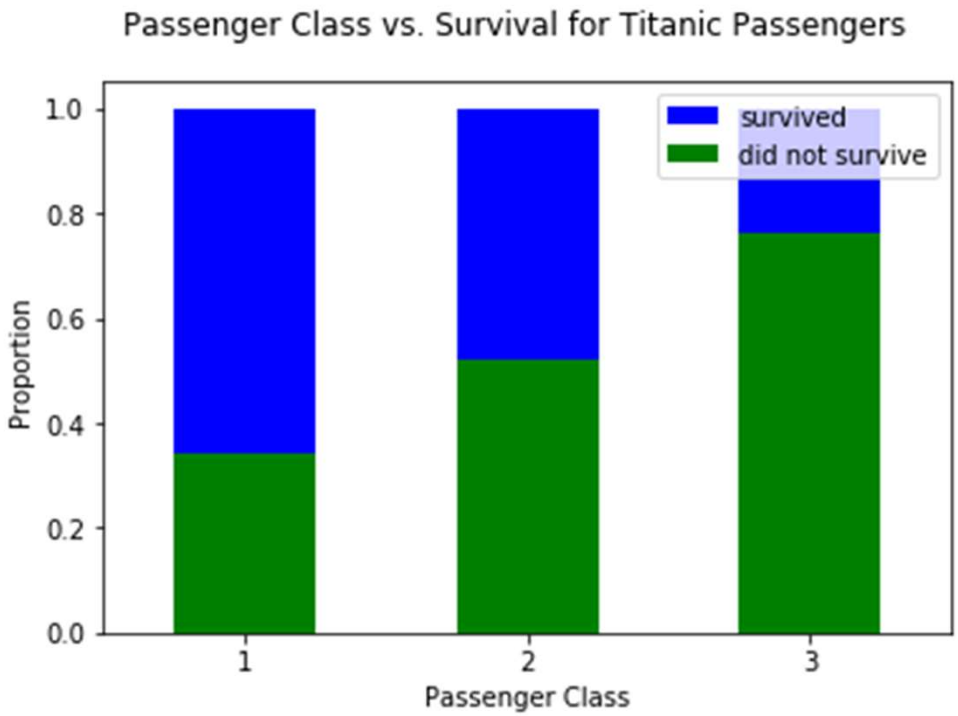
*for each passenger class, **how many** passengers survived vs. did not survive*?

Categorical vs Categorical



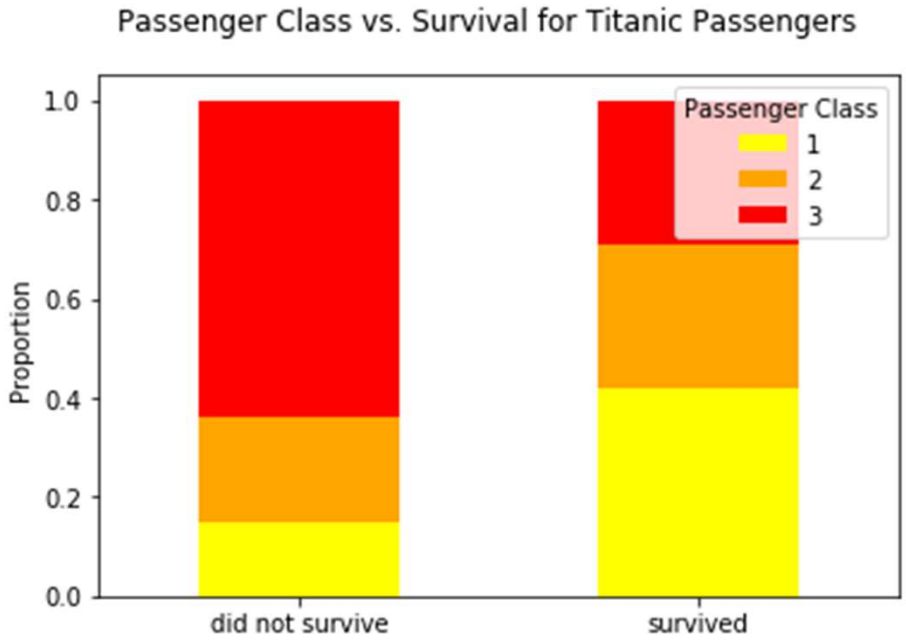
how many *were in each passenger class*?

Categorical vs Categorical



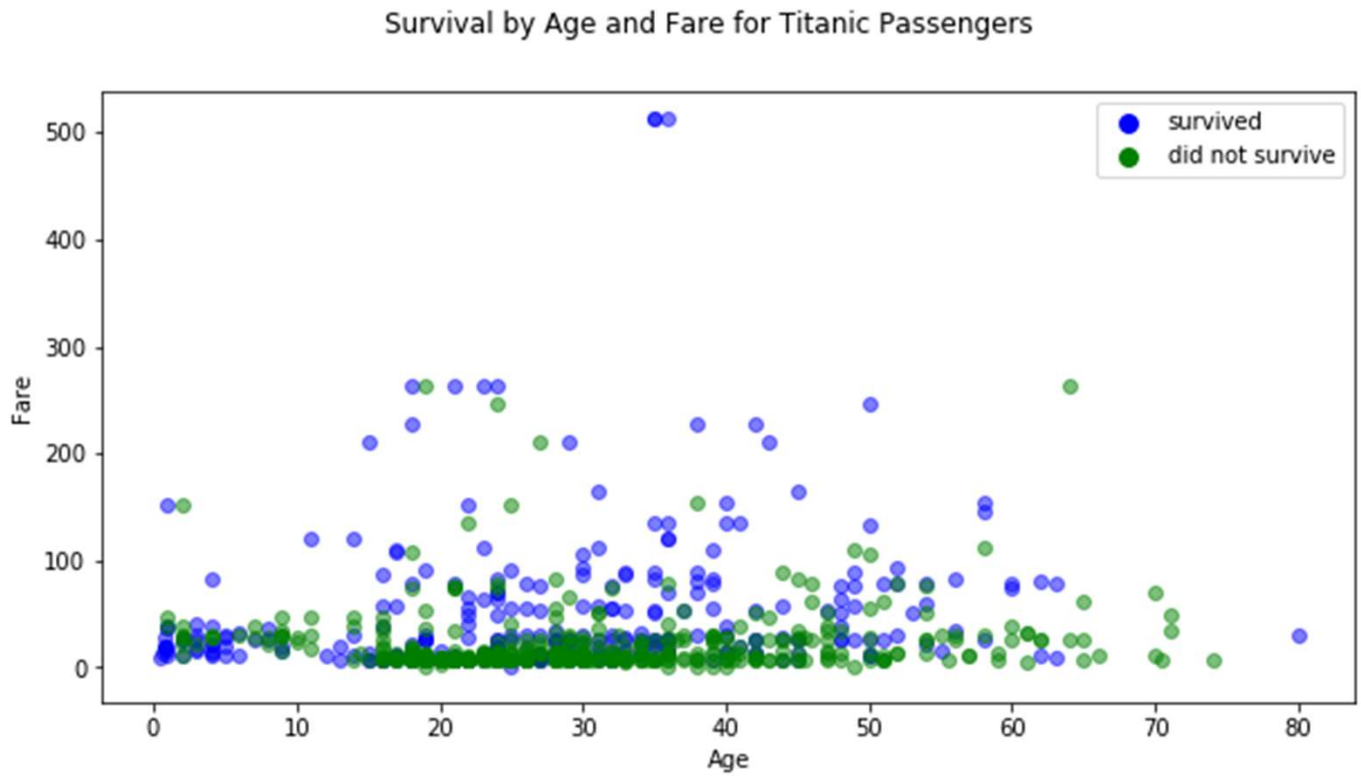
*for each passenger class, **what proportion** of passengers survived vs. did not survive*?

Categorical vs Categorical

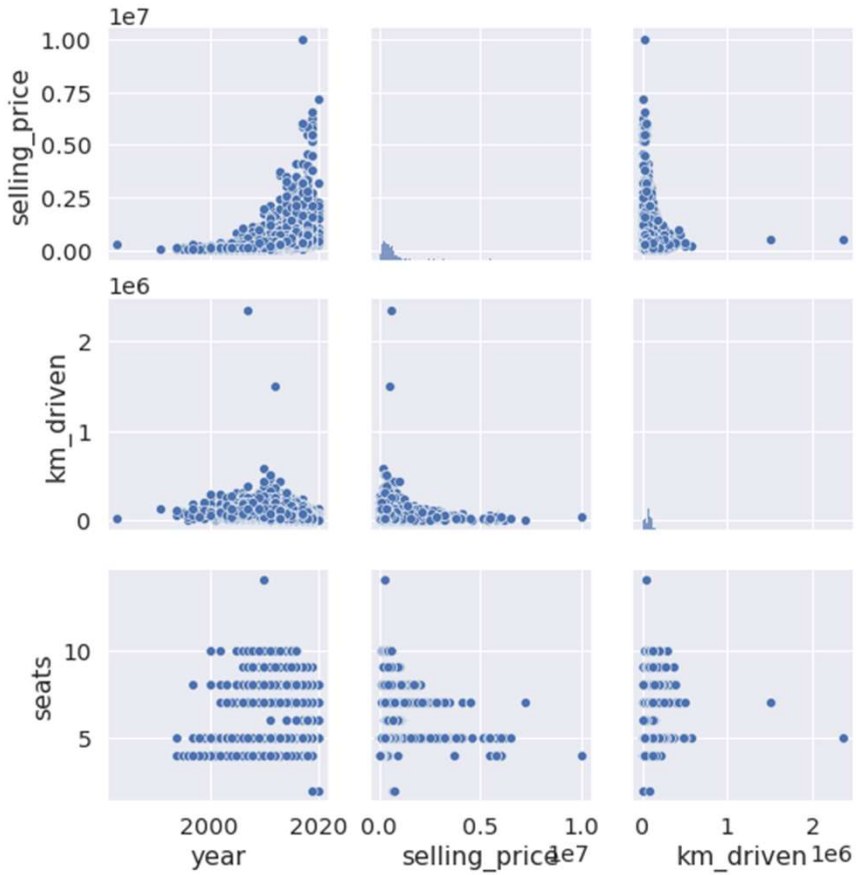


for passengers who survived vs. did not survive, ***what proportion*** *were in each passenger class*?

Numeric vs. Numeric vs. Categorical



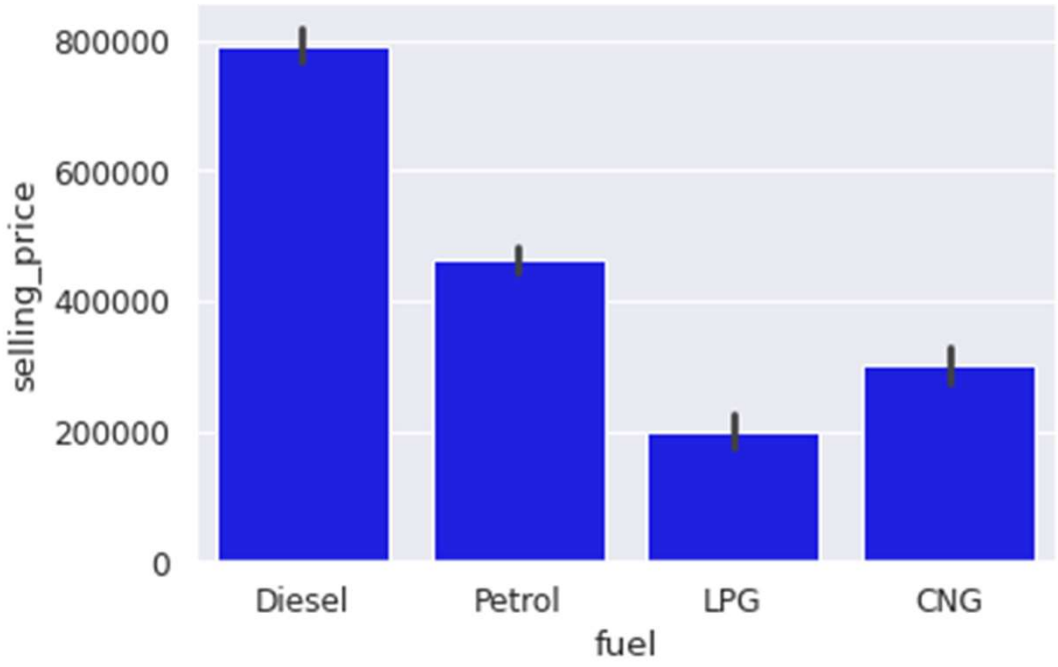
Numeric vs. Numeric



Pair plot

categorical and numeric

```
sns.barplot(
x='fuel',
y='selling_price',
data=cars,
color='blue',
# estimator=sum,
# estimator=np.median);
```



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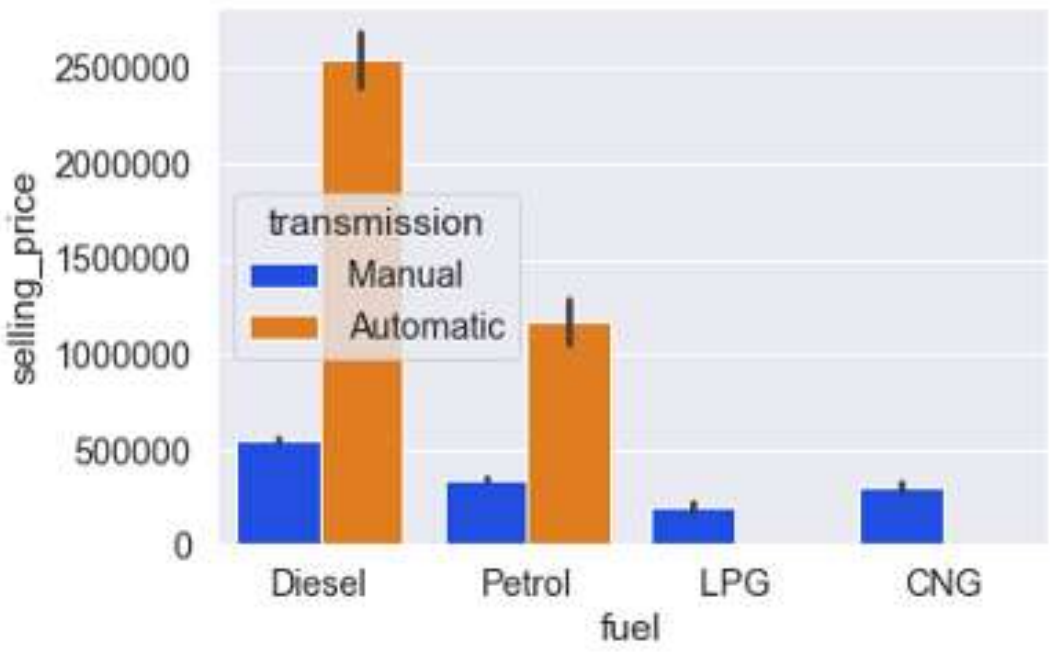
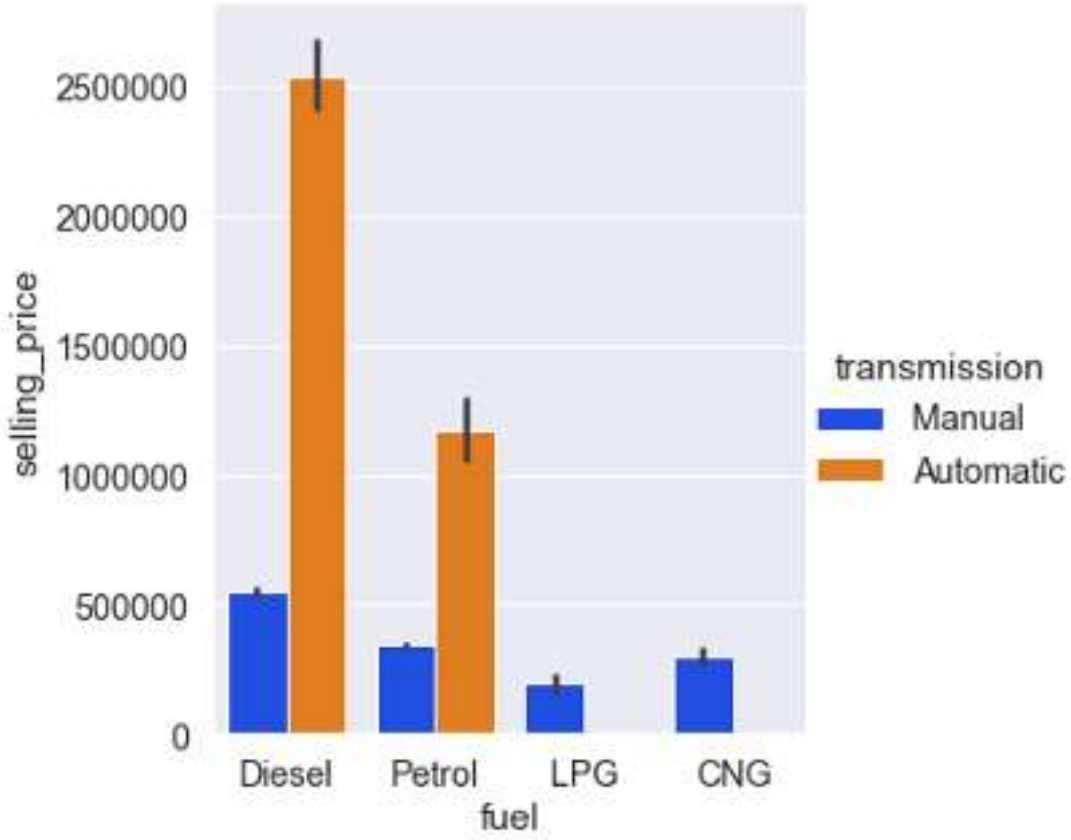
```
sns.barplot(
    x='fuel',
    y='selling_price',
    data=cars,
    palette='bright',
    hue='transmission');
```

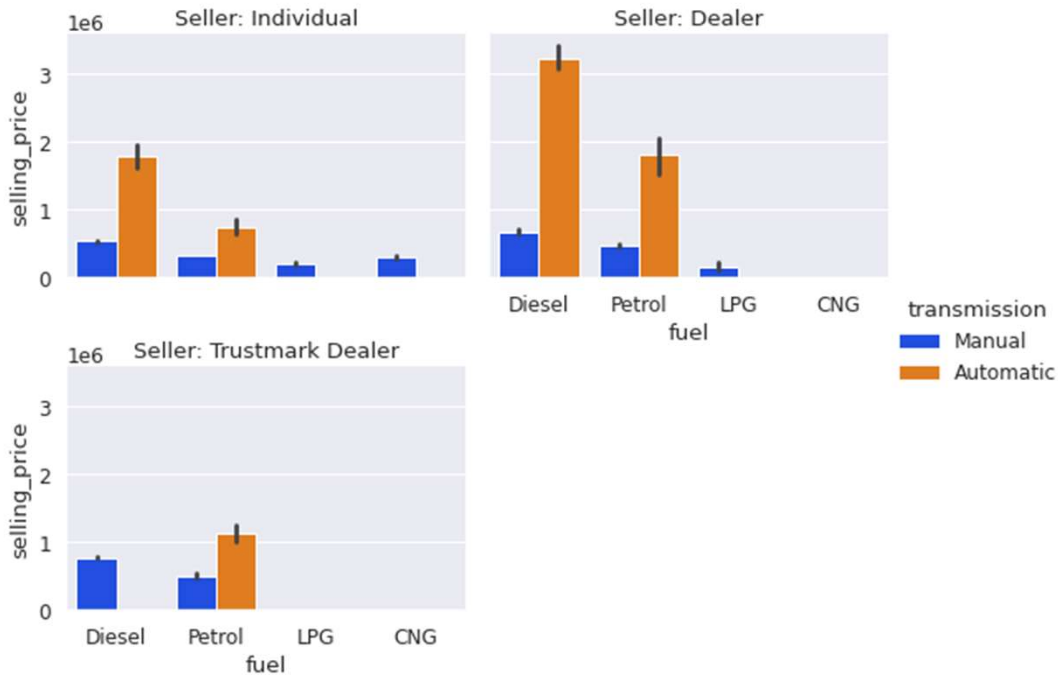


```
sns.catplot(
x='fuel',
y='selling_price',
data=cars,
palette='bright',
kind='bar',
hue='transmission');
```



catplot

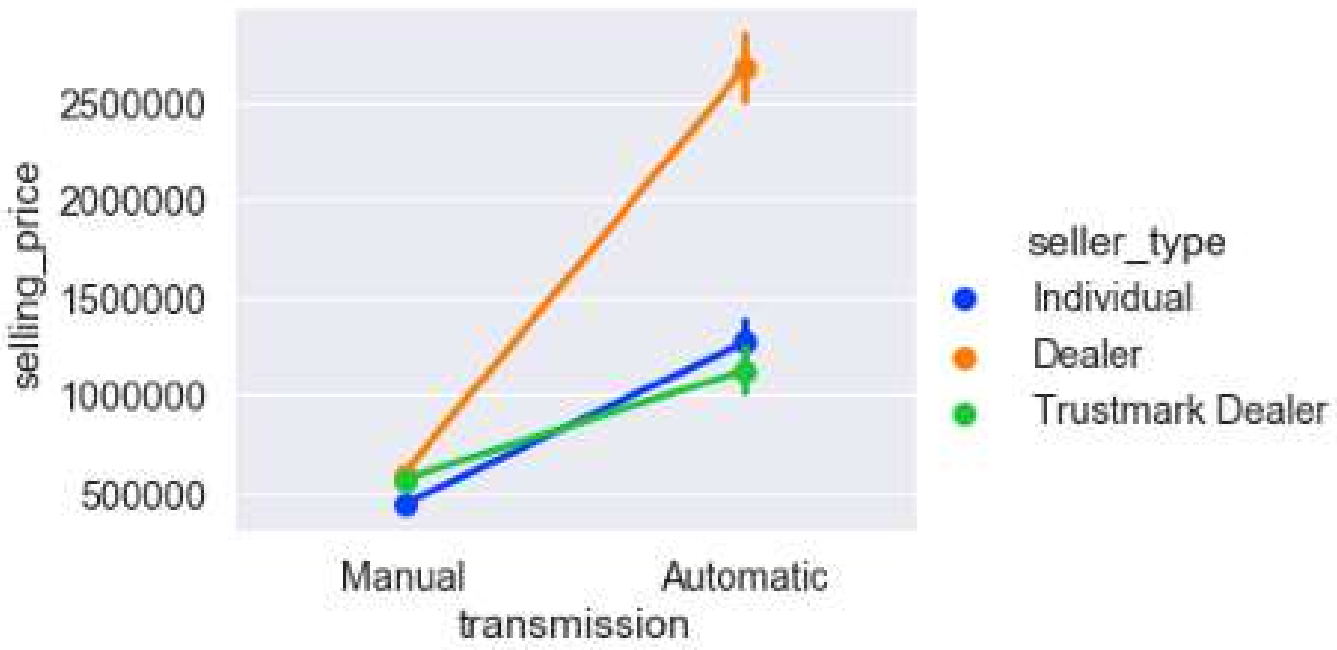
```
g = sns.catplot(
        x='fuel',
        y='selling_price',
        data=cars,
        palette='bright',
        height=3, aspect=1.3,
        kind='bar',
        hue='transmission',
        col ='seller_type',
        col_wrap=2)
g.set_titles(
    'Seller: {col_name}');
```



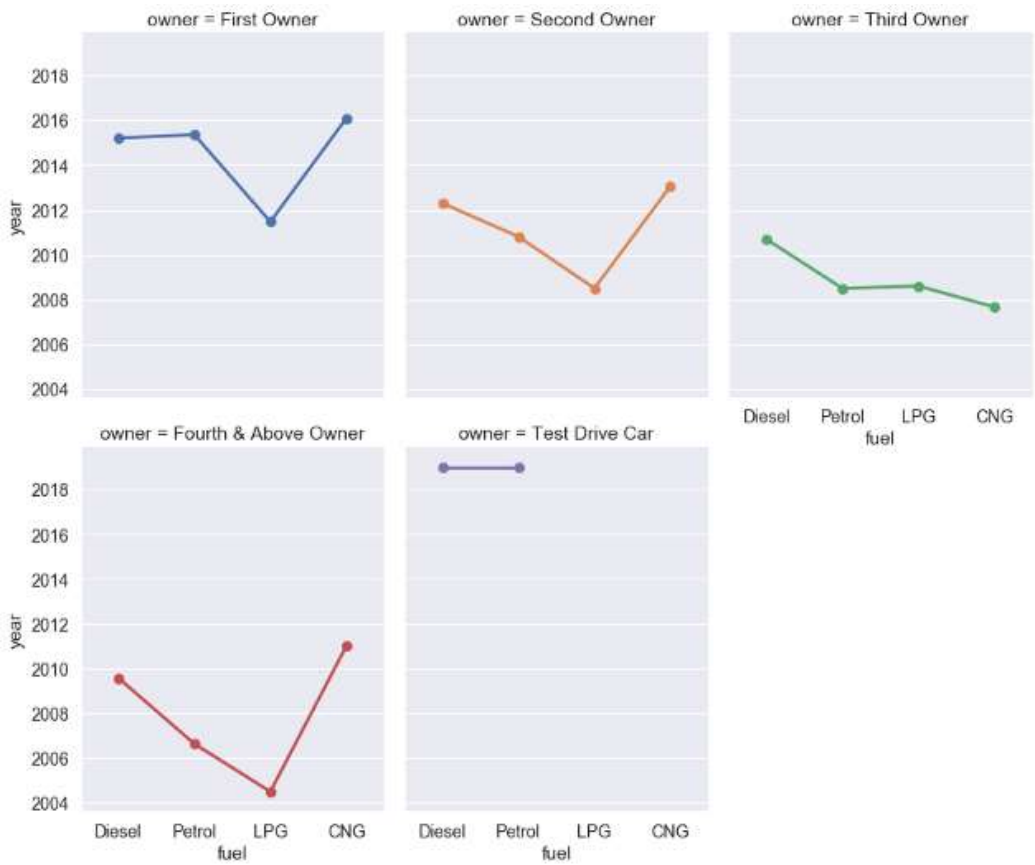
Point plot

```
sns.pointplot(
x='seller_type',
y='mileage_kmpl',
data=cars);
```

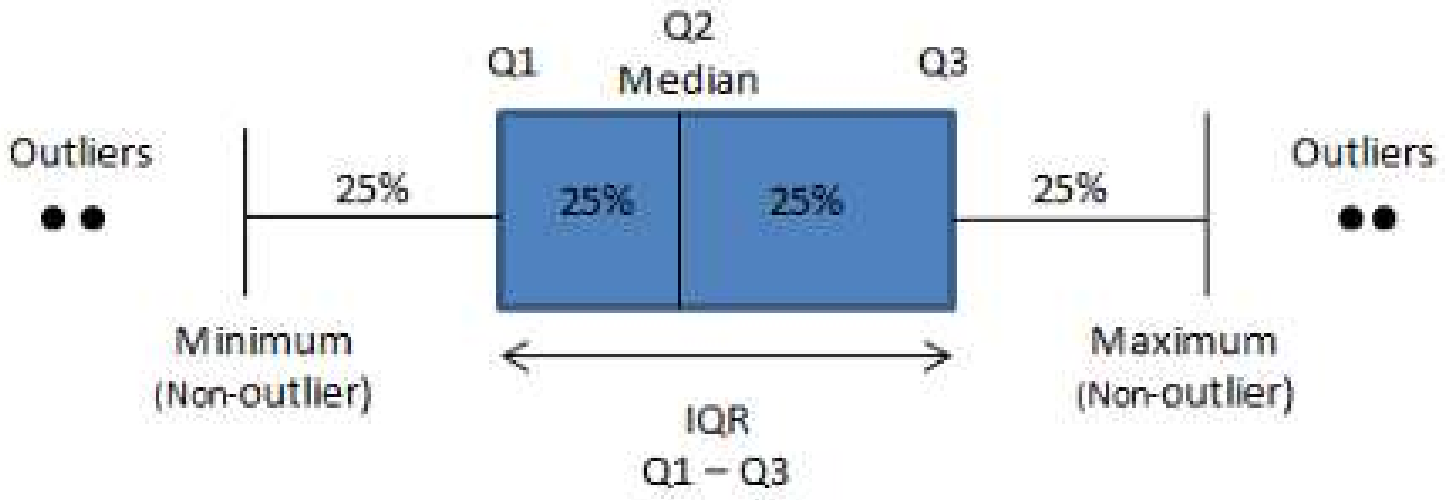
```
sns.catplot(
x='transmission',
y='selling_price',
data=cars,
palette='bright',
kind='point',
hue='seller_type');
```



```
sns.catplot(
    x='fuel',
    y='year',
    data=cars,
    ci=None,
    height=5, #default
    aspect=.8,
    kind='point',
    hue='owner',
    col='owner',
    col_wrap=3);
```

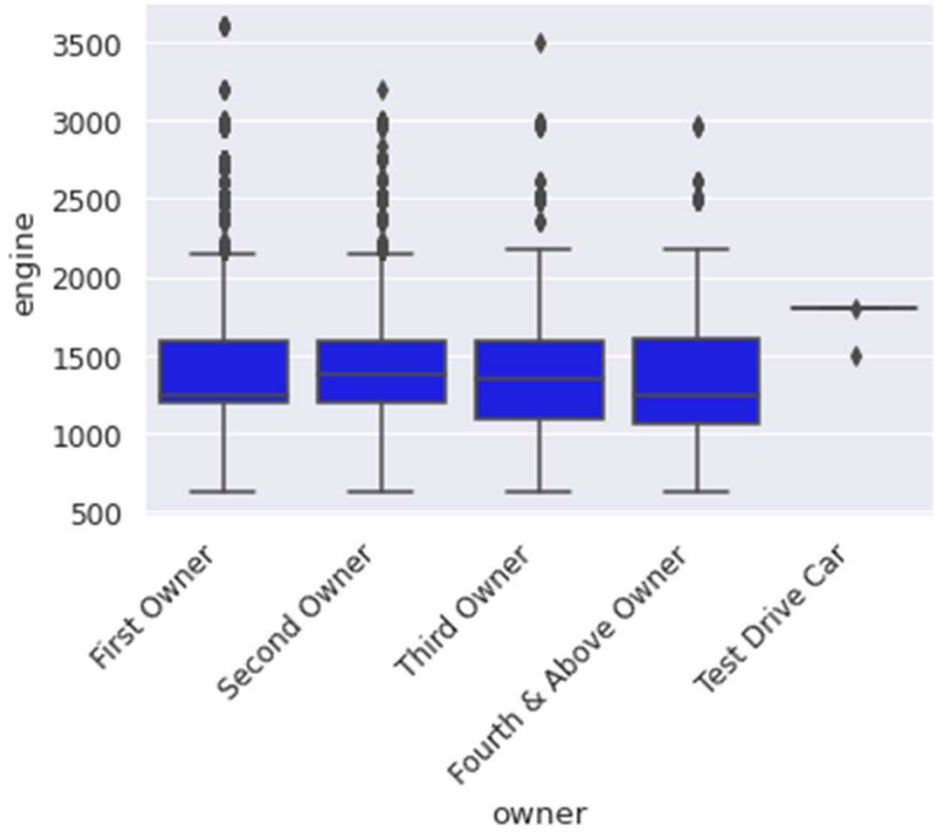


Box Plot

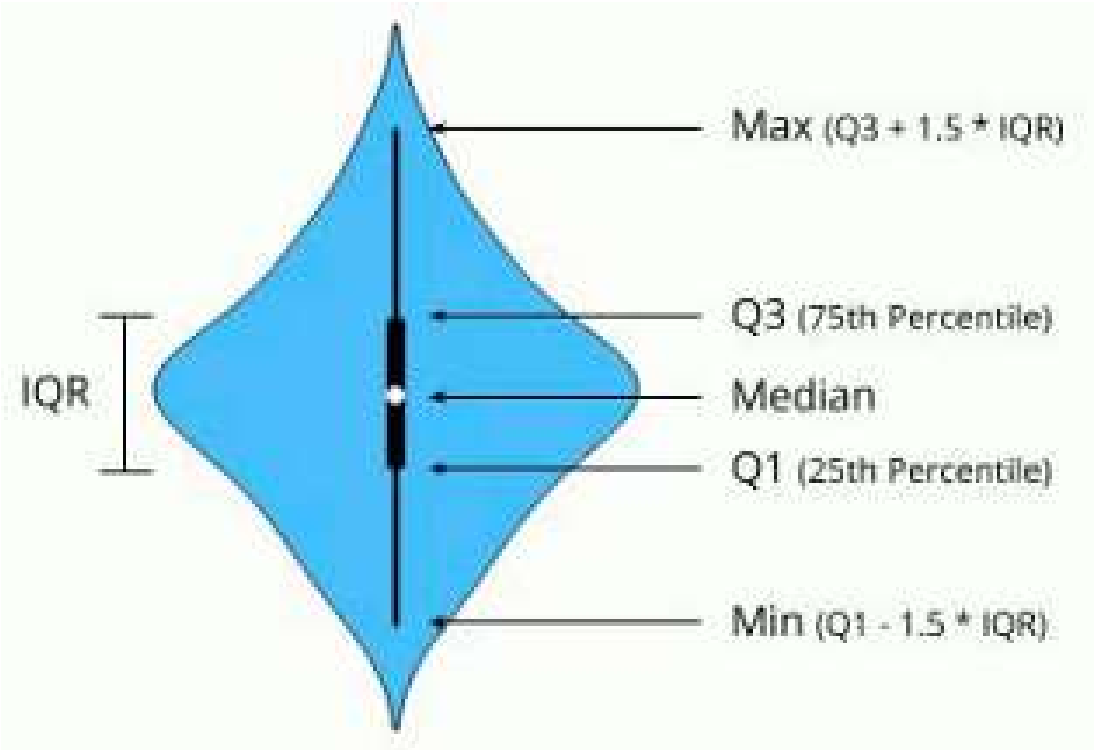

Q2
Median
Q1
Q3
Outliers
25%
25%
25%
25%
Outliers
Minimum
(Non-outlier)
Maximum
(Non-outlier)
IQR
Q1 – Q3

Boxplot

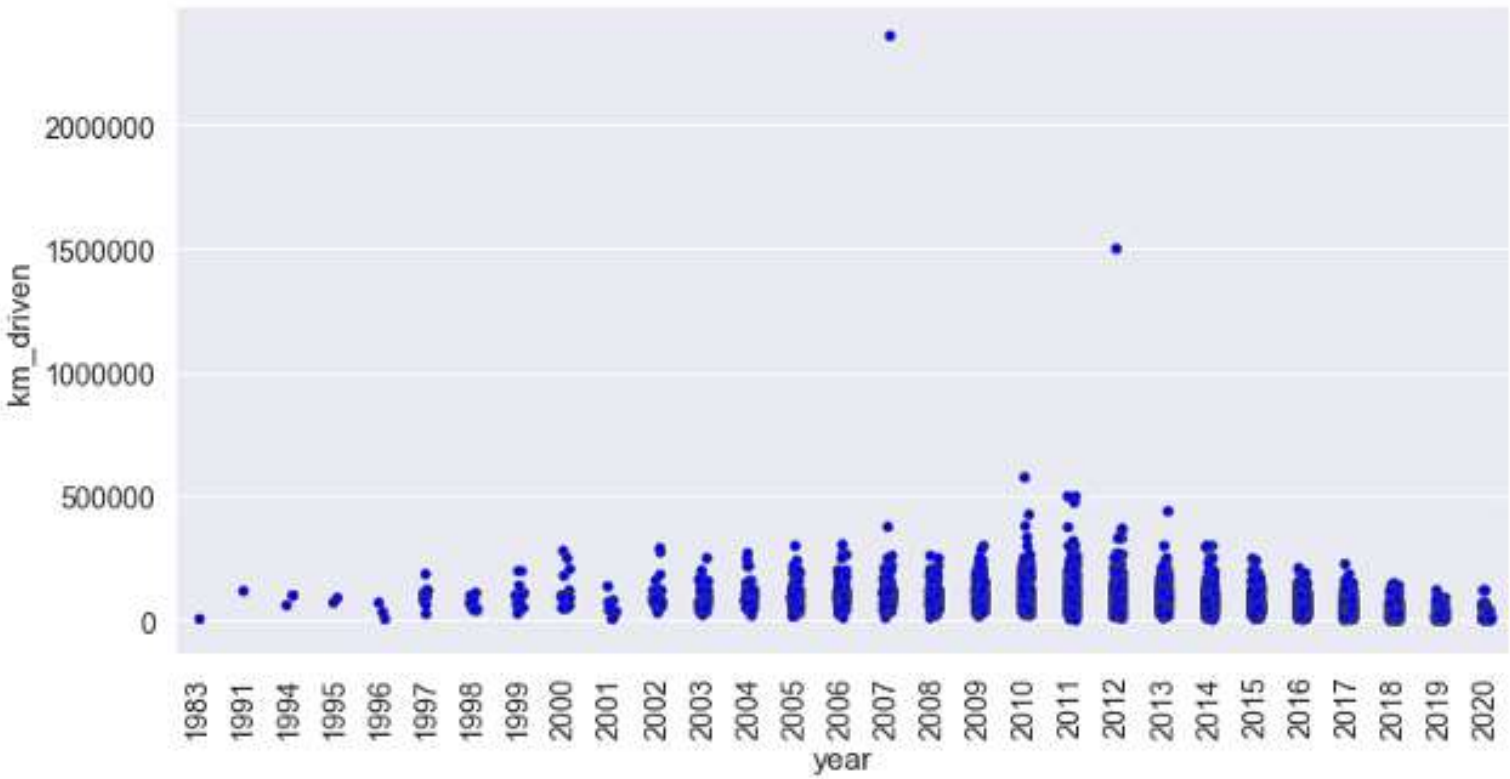
```
sns.boxplot(
    x='owner',
    y='engine',
    data=cars,
    color='blue')
plt.xticks(rotation=45,
           ha='right');
```



Violin plot


Max (Q3 + 1.5 * IQR)
Q3 (75th Percentile)
IQR
Median
Q1 (25th Percentile)
Min (Q1 - 1.5 * IQR)

Strip plot


2000000
1500000
1000000
500000
0
km_driven
1983
1991
1994
1995
1996
1997
1998
1999
2000
2001
2002
2003
2004
2005
2006
2007
2008
2009
2010
2011
2012
2013
2014
2015
2016
2017
2018
2019
2020
year

IN
GOD
WE
TRUST

References

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- https://towardsdatascience.com/10-must-know-seaborn-functions-for-multivariate-data-analysis-in-python-7ba94847b117
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